

STUDENT PROJECT ASSIGNMENT4

Title

Trim, Thrust Required, and Power Required Analysis of a Cropped Delta Wing UAV Using MATLAB / Excell.

Project Type

Computational + Analytical Project
(Aircraft Performance & Flight Mechanics)

Objective

The objective of this project is to analyse the trim condition, thrust required, and power required of a cropped delta wing UAV in steady level flight, using given aerodynamic and geometric parameters. Students will develop a MATLAB-based iterative procedure to evaluate how flight velocity, thrust required, power required, and performance metrics vary with angle of attack.

Background and Motivation

Aircraft performance analysis requires understanding how aerodynamic forces, moments, and propulsion requirements vary with flight condition. For unconventional configurations such as cropped delta reflex wings, analytical solutions are not straightforward and must be evaluated using numerical and iterative methods.

In this project, students apply trim equations, drag polar modeling, and level-flight equilibrium to compute:

- Trim elevator deflection
- Flight velocity
- Lift and drag coefficients
- Thrust required
- Power required

The project bridges **aerodynamics, flight mechanics, and numerical programming.**

UAV Configuration and Given Data

Geometry

- Configuration: **Cropped delta wing (wing-alone)**
- Root chord, (C_r) = 0.9 m
- Tip chord, (C_t) = 0.15 m
- Span, (b) = 1.5 m
- Oswald efficiency factor, (e) = 0.89

Aerodynamic Coefficients

$$C_{D0} = 0.03$$

$$C_{m0} = 0.01$$

$$C_{L\alpha} = 2.92 \text{ per rad}$$

$$C_{L\delta_e} = 0.265$$

$$C_{m\alpha} = -0.292$$

$$C_{m\delta_e} = -0.4$$

Flight Conditions

- UAV mass = 3.5 kg
- Flight condition: **Steady level flight**
- Air density (sea level): $\rho = 1.225 \text{ kg/m}^3$
- Gravity: $g = 10 \text{ m/s}^2$

Preprocessing Calculations

Students must first compute the following geometric parameters:

1. Taper Ratio

$$\lambda = \frac{C_t}{C_r}$$

2. Planform Area

$$S = \frac{b}{2} C_r (1 + \lambda)$$

3. Aspect Ratio

$$AR = \frac{b^2}{S}$$

4. Induced Drag Correction Factor

$$k = \frac{1}{\pi e AR}$$

Trim and Performance Theory

Trim Conditions

For steady, trimmed flight:

$$\begin{aligned} C_m &= C_{m0} + C_{m\alpha}\alpha + C_{m\delta_e}\delta_e = 0 \\ C_L &= C_{L0} + C_{L\alpha}\alpha + C_{L\delta_e}\delta_e \end{aligned}$$

From these equations, students will compute:

- Trim elevator deflection $\delta_{e,\text{trim}}$
- Corresponding lift coefficient $C_{L,\text{trim}}$

Drag Polar

$$C_D = C_{D0} + kC_L^2$$

Level Flight Relations

- Lift equilibrium:

$$L = W = \frac{1}{2} \rho V^2 S C_L$$

- Thrust required:

$$T_r = D = \frac{1}{2} \rho V^2 S C_D$$

- Power required:

$$P_r = T_r V$$

Project Tasks

Phase 1 – MATLAB Model Development

Students must develop a MATLAB code to:

1. Vary angle of attack:

$$\alpha = 0^\circ \text{ to } 12^\circ (\Delta\alpha = 0.5^\circ)$$

2. For each angle of attack:
 - a. Compute trim elevator deflection
 - b. Compute lift coefficient
 - c. Determine flight velocity
 - d. Compute drag coefficient
 - e. Calculate thrust required
 - f. Calculate power required

Phase 2 – Performance Metrics

Compute and store:

- C_L/C_D
- $C_L^{3/2}/C_D$

Phase 3 – Visualization

Generate plots showing variation of the following with **flight velocity or angle of attack**:

1. Trim angle of attack
2. Trim elevator deflection
3. Lift coefficient
4. Drag coefficient
5. Thrust required
6. Power required
7. C_L/C_D
8. $C_L^{3/2}/C_D$

Deliverables

Each student group must submit:

1. **Project Report** including:
 - a. Problem description
 - b. Geometry and aerodynamic modeling
 - c. Trim equations and assumptions
 - d. MATLAB algorithm
 - e. Results and discussion
 - f. Key performance trends
2. **MATLAB Code / Excell sheet**
 - o Well-commented
 - o Modular and readable
3. **Plots and Figures**
 - o Clearly labelled axes
 - o Units included
4. **Evaluation Criteria**

Component	Weightage
Theory & Formulation	25%
MATLAB Implementation	30%
Results & Interpretation	25%
Report Quality	10%

Learning Outcomes

After completing this project, students will be able to:

- Perform **trim analysis** of a UAV
- Relate **angle of attack to flight velocity**
- Compute **thrust and power requirements**
- Understand performance metrics such as C_L/C_D and $C_L^{3/2}/C_D$
- Develop **iterative MATLAB solvers** for flight mechanics problems

Concluding Remark for Students

This project emphasizes that **aircraft performance is governed by coupled aerodynamic, propulsive, and trim constraints**, and that **numerical methods are essential** for analyzing real UAV configurations.